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Aim: Performing Data Analysis using Calculations in Tableau

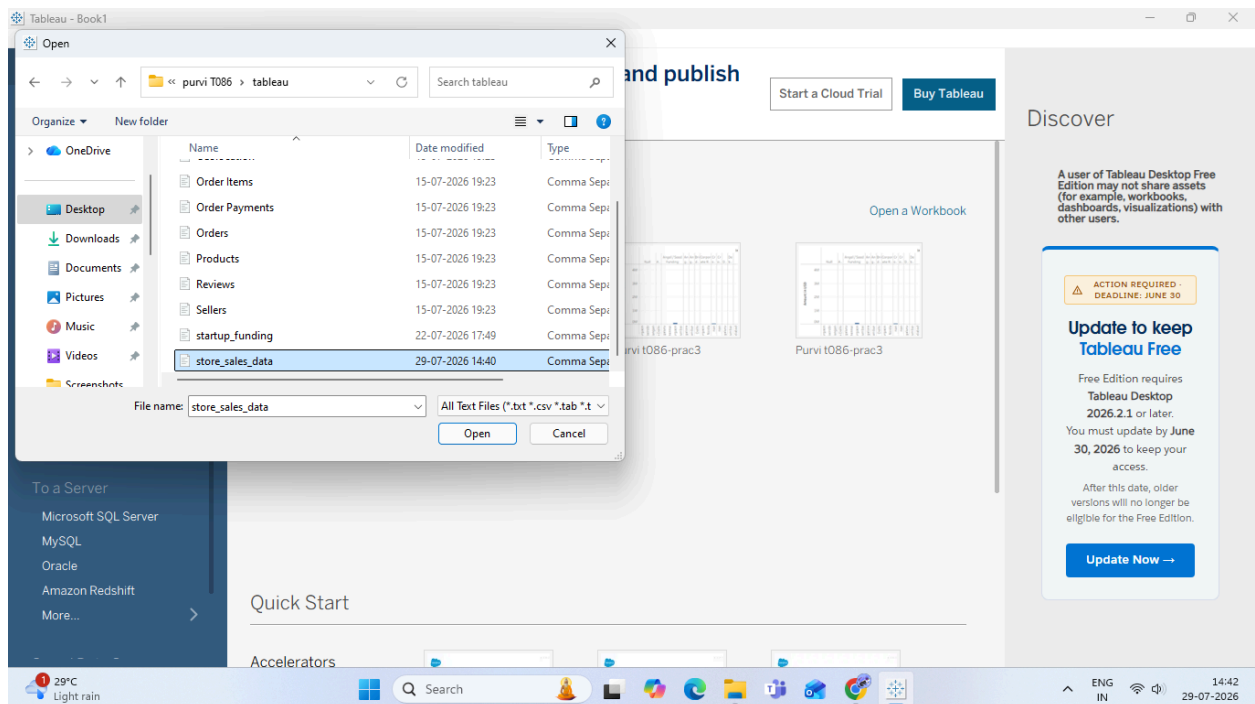
- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights



A: Creating Basic Calculated Fields Using Formulas

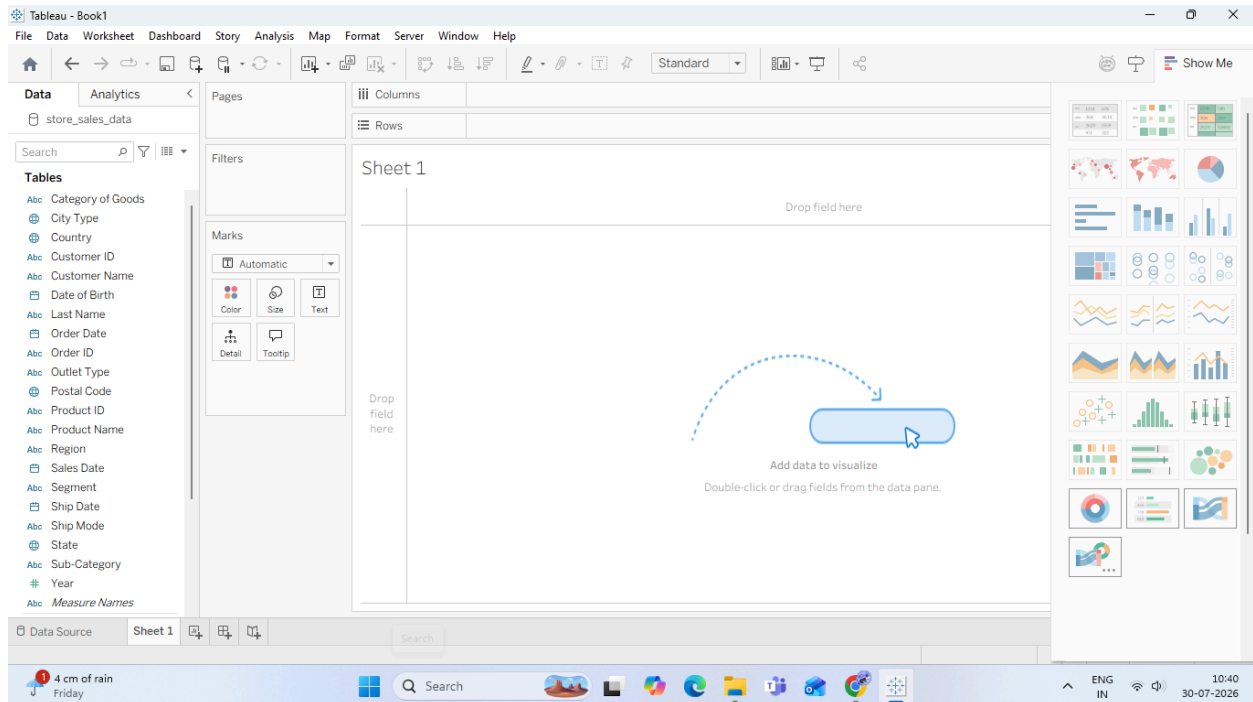
1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your store_sales_data.csv, and click Open.



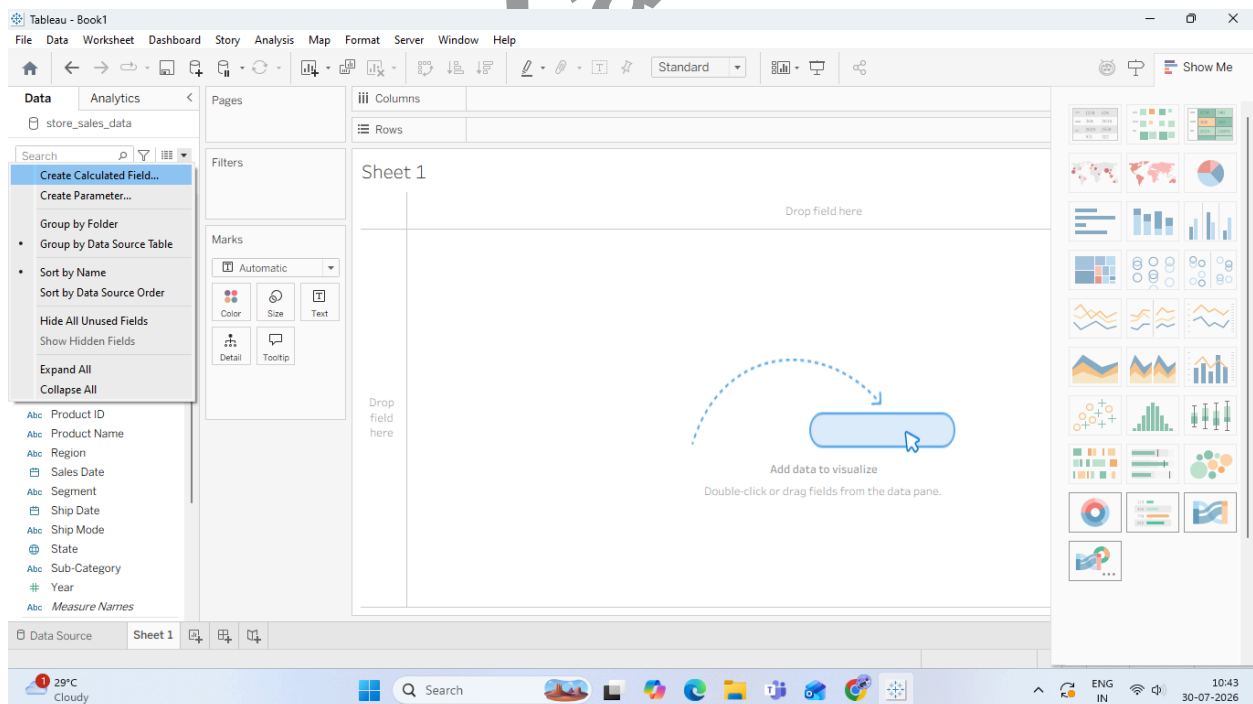
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2. Click on Sheet 1 at the bottom-left corner to enter your workspace.



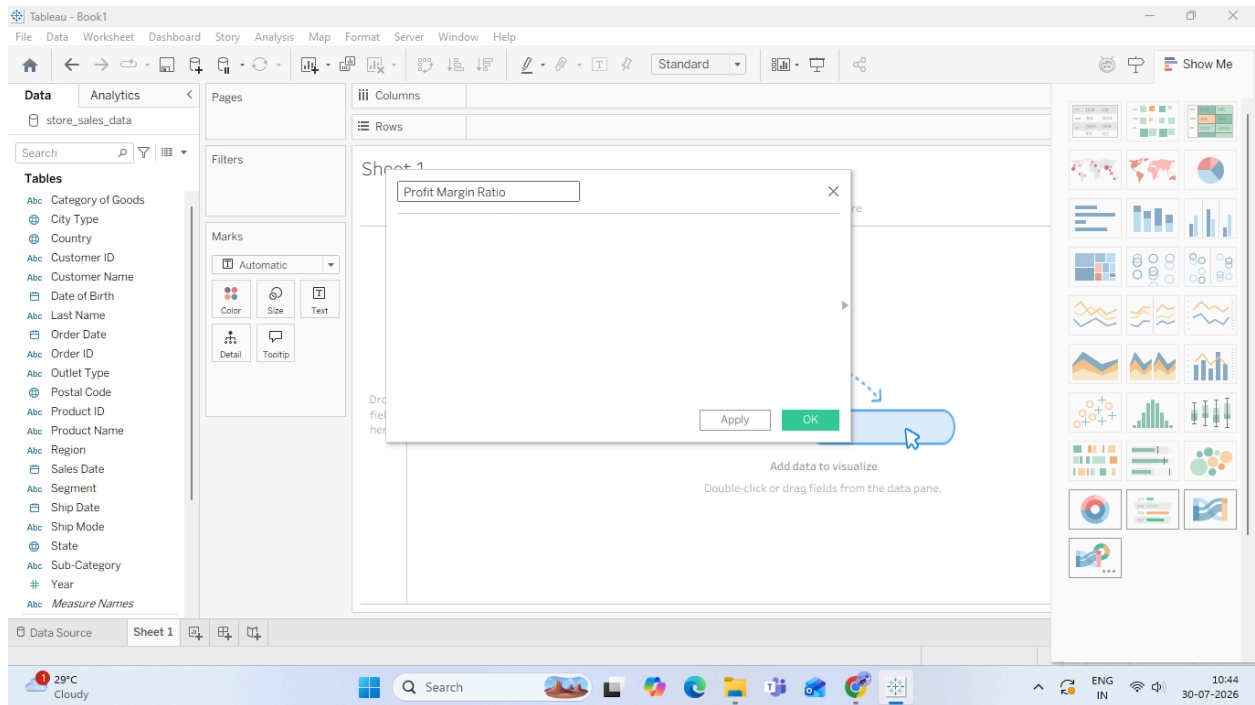
2. Building a Basic Custom Calculation

1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane. Create Calculated Field

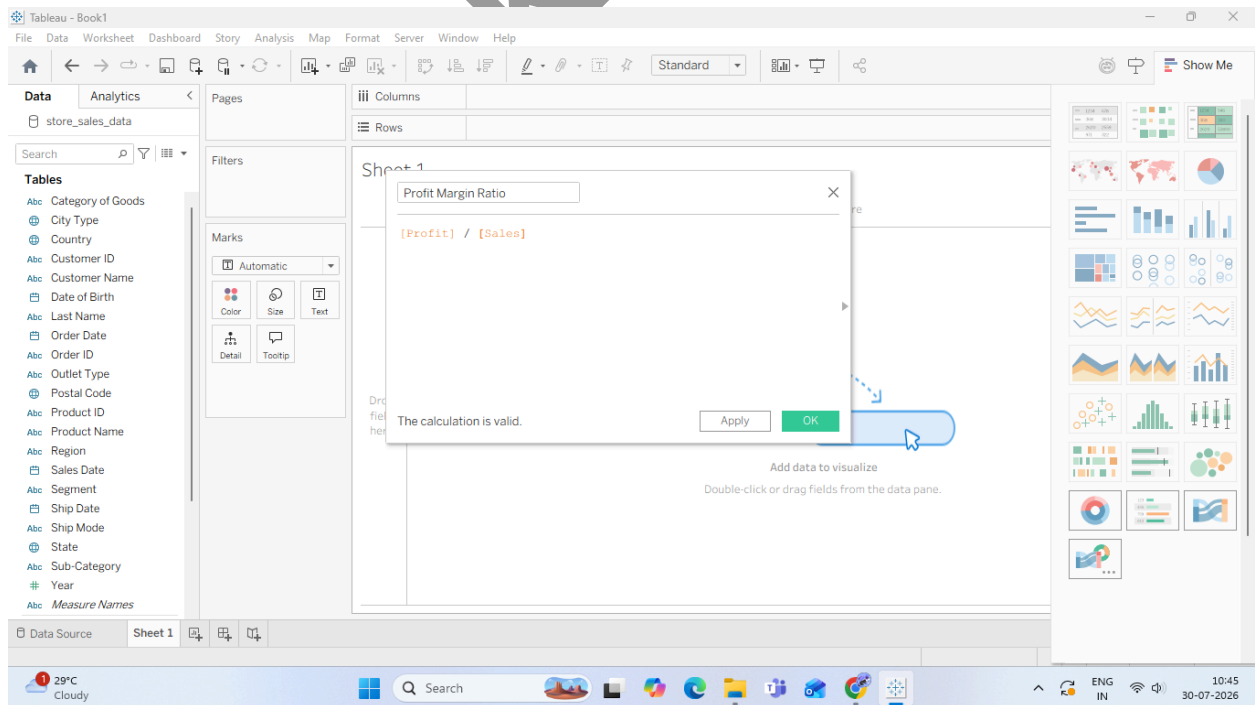


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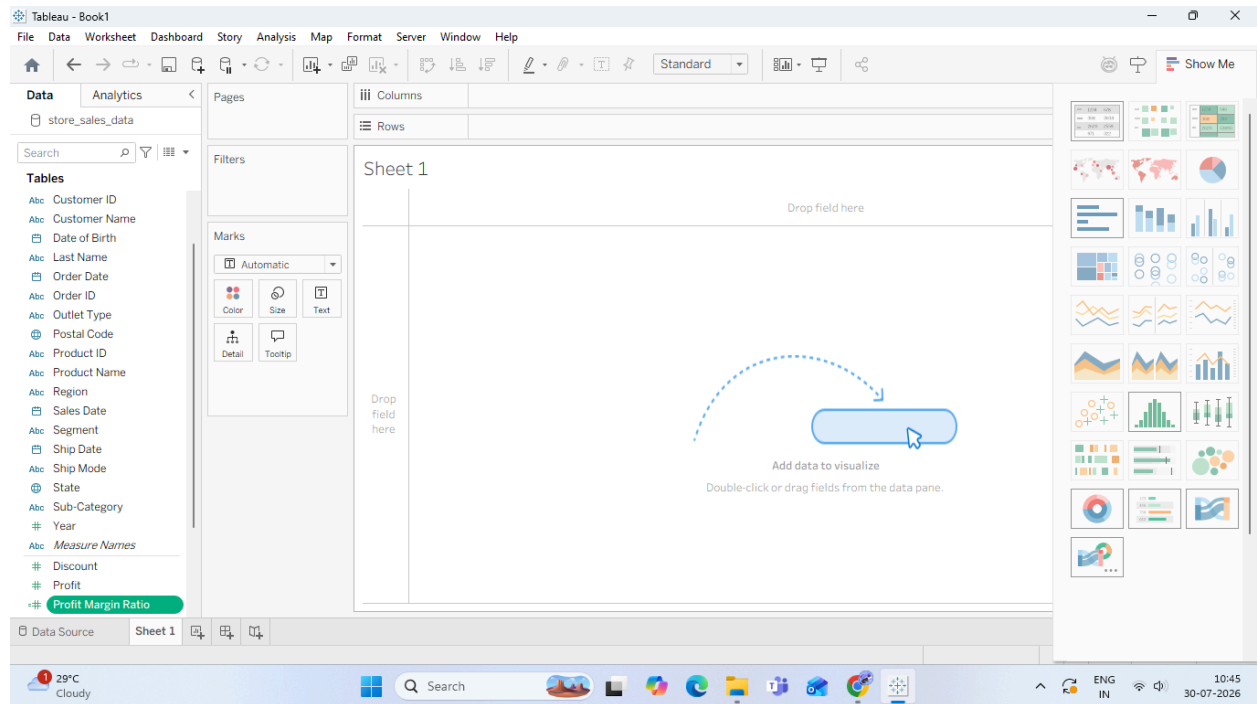
2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.



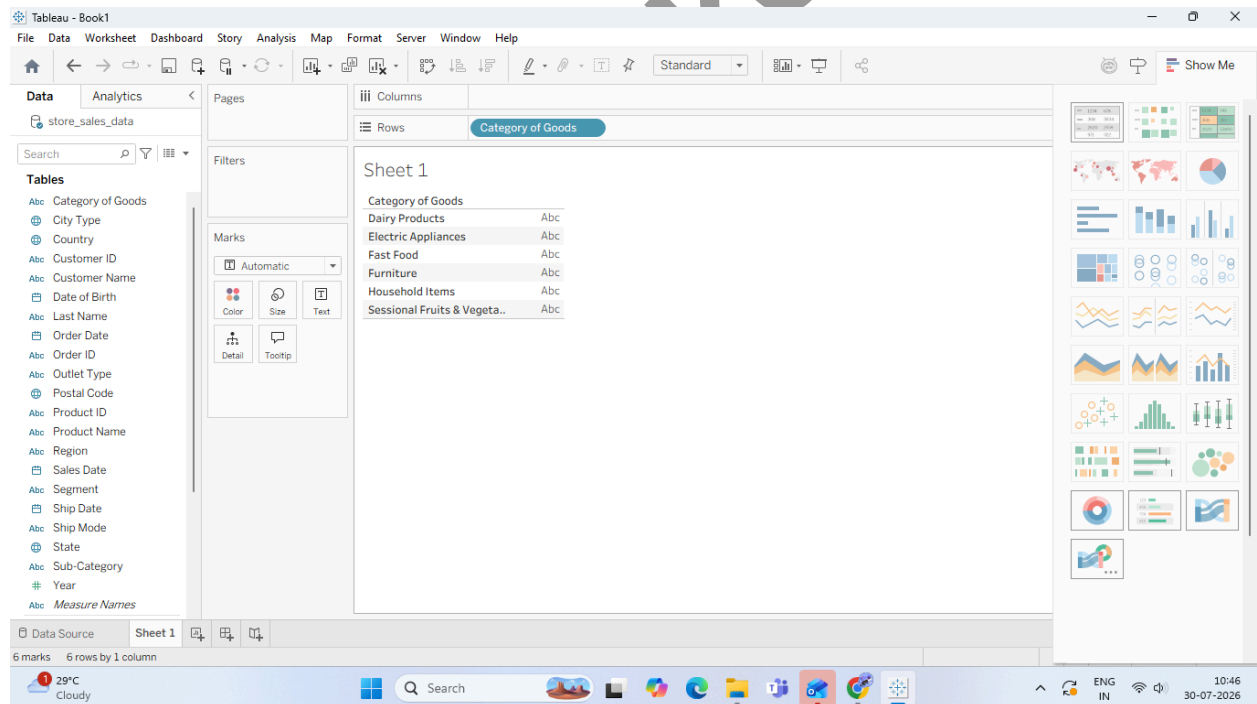
3. In the main workspace box, enter the following basic mathematical formula:
[Profit] / [Sales]



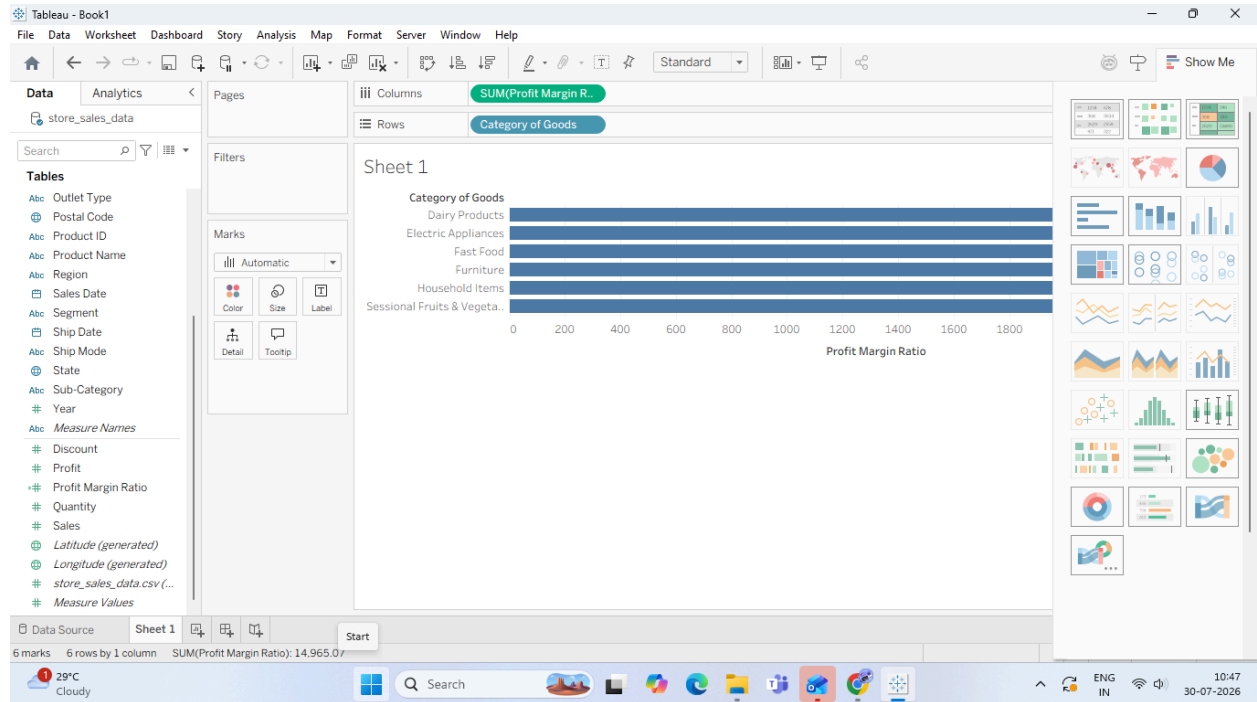
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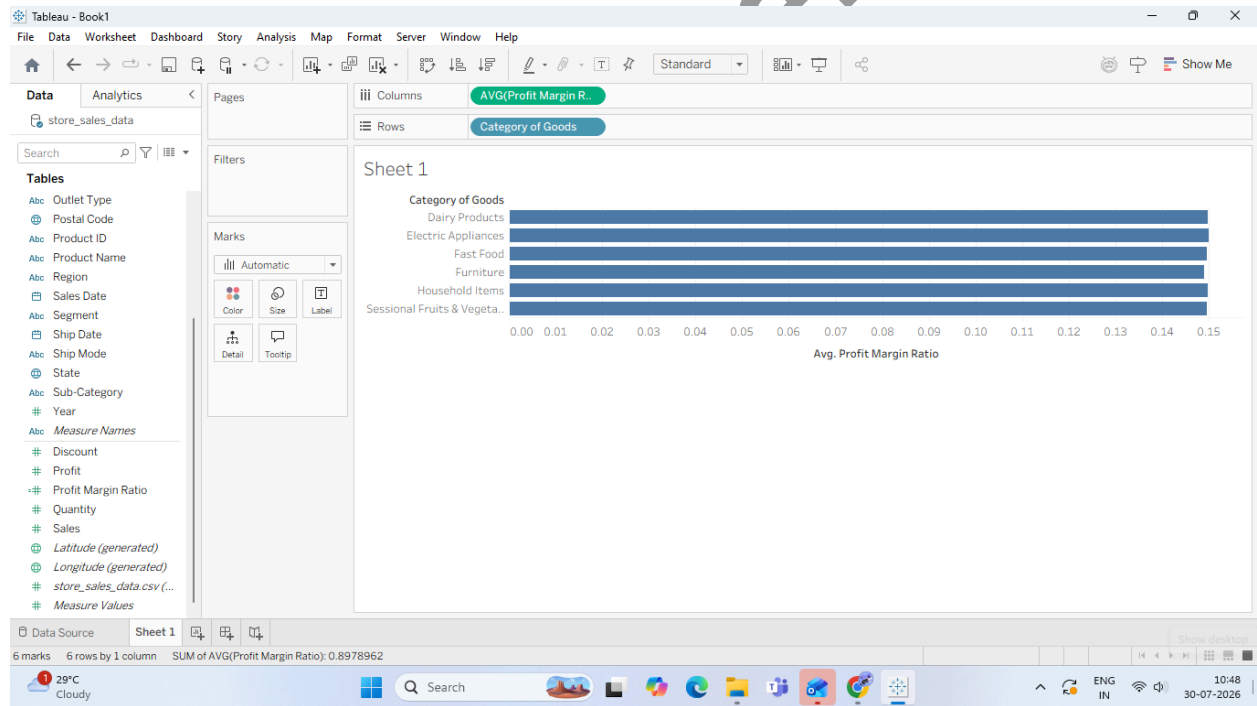
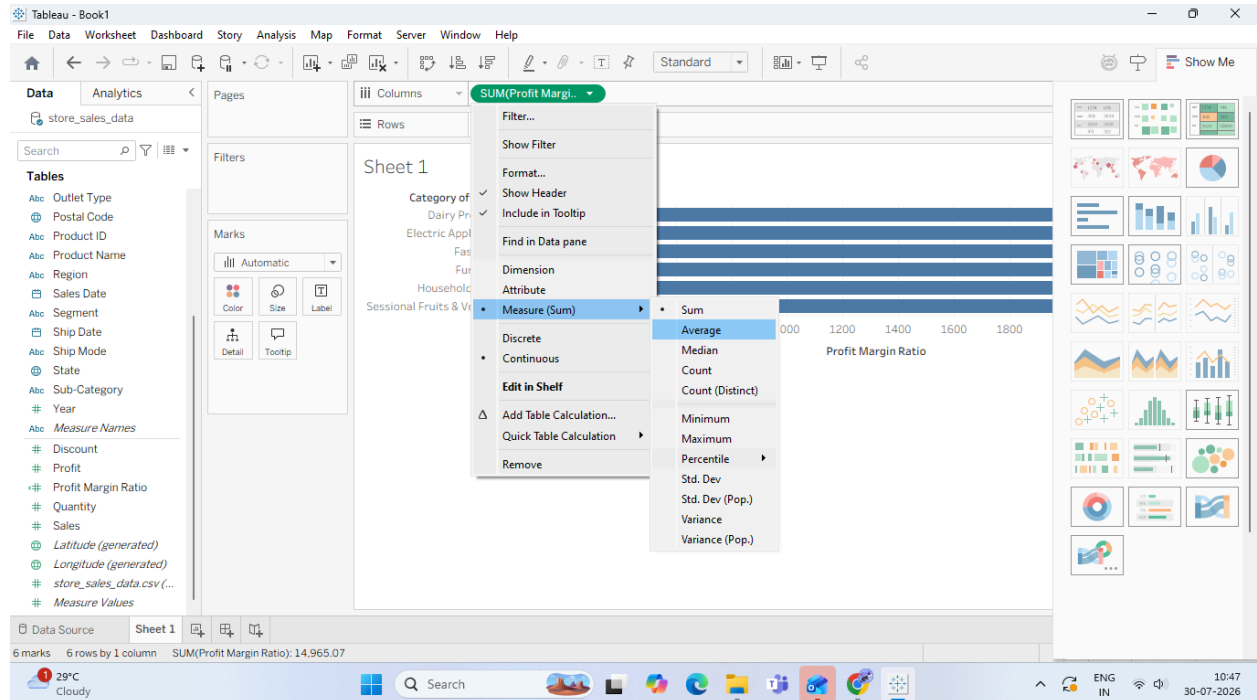
2. Putting the Field to Use



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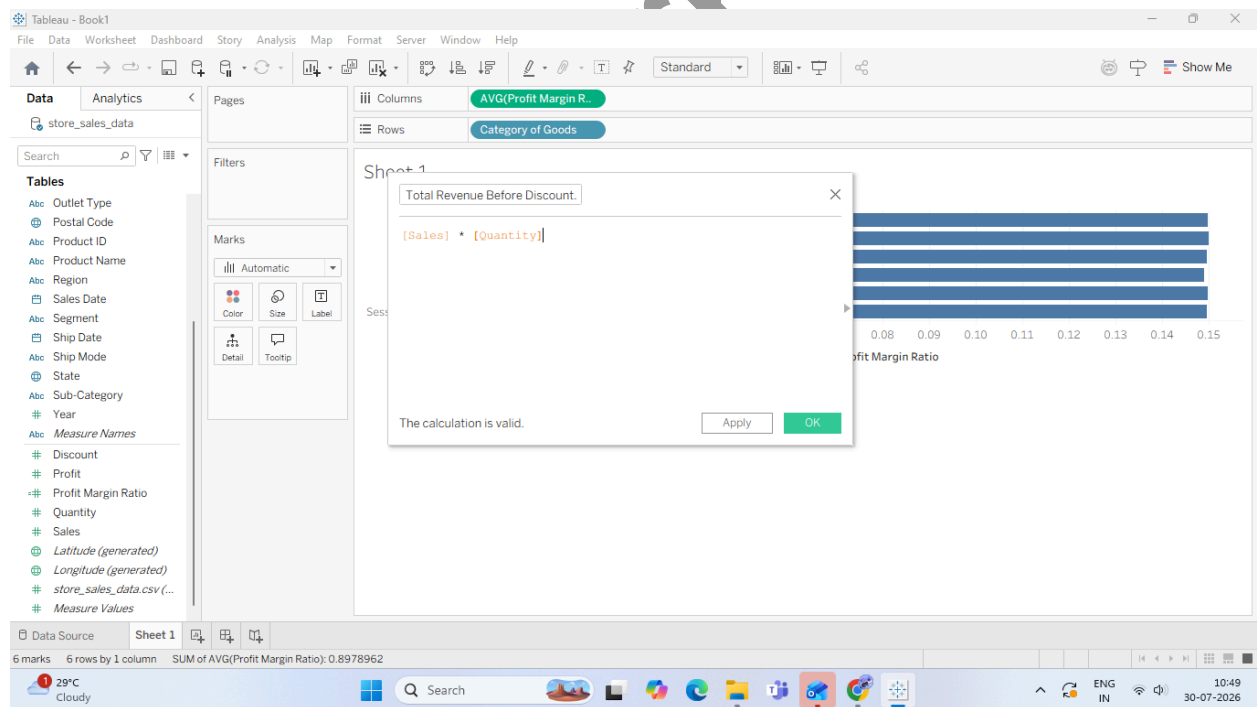
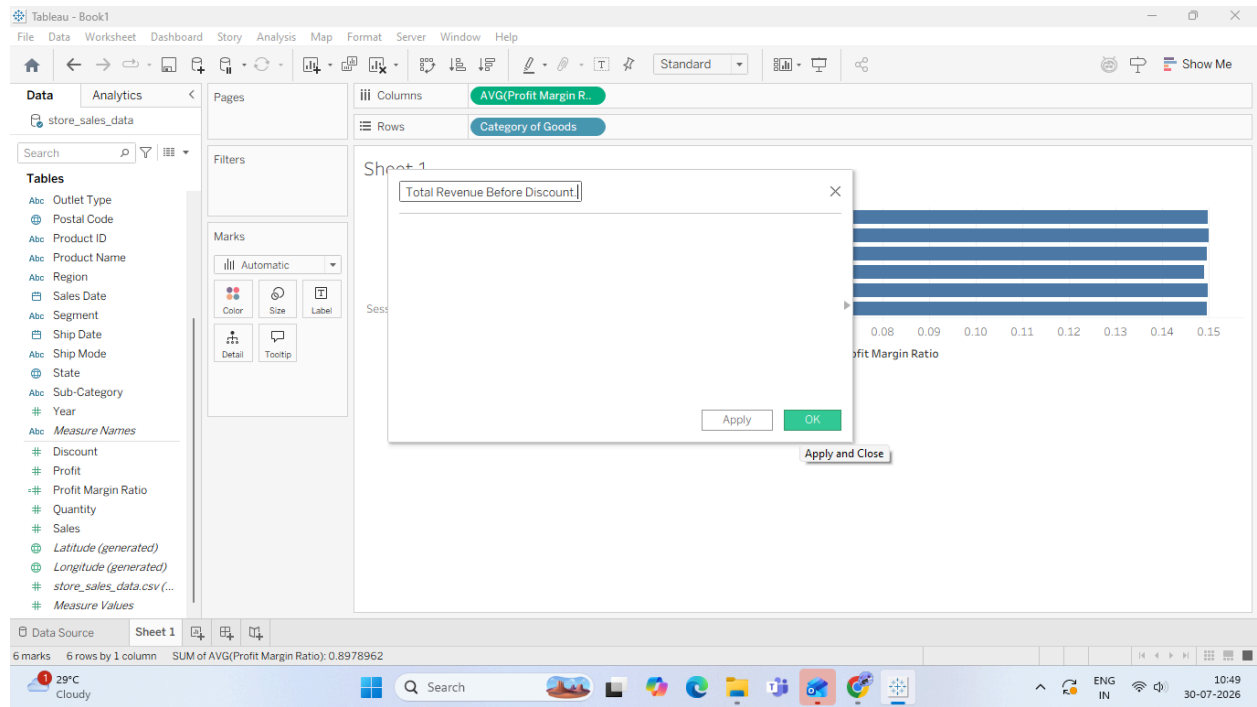
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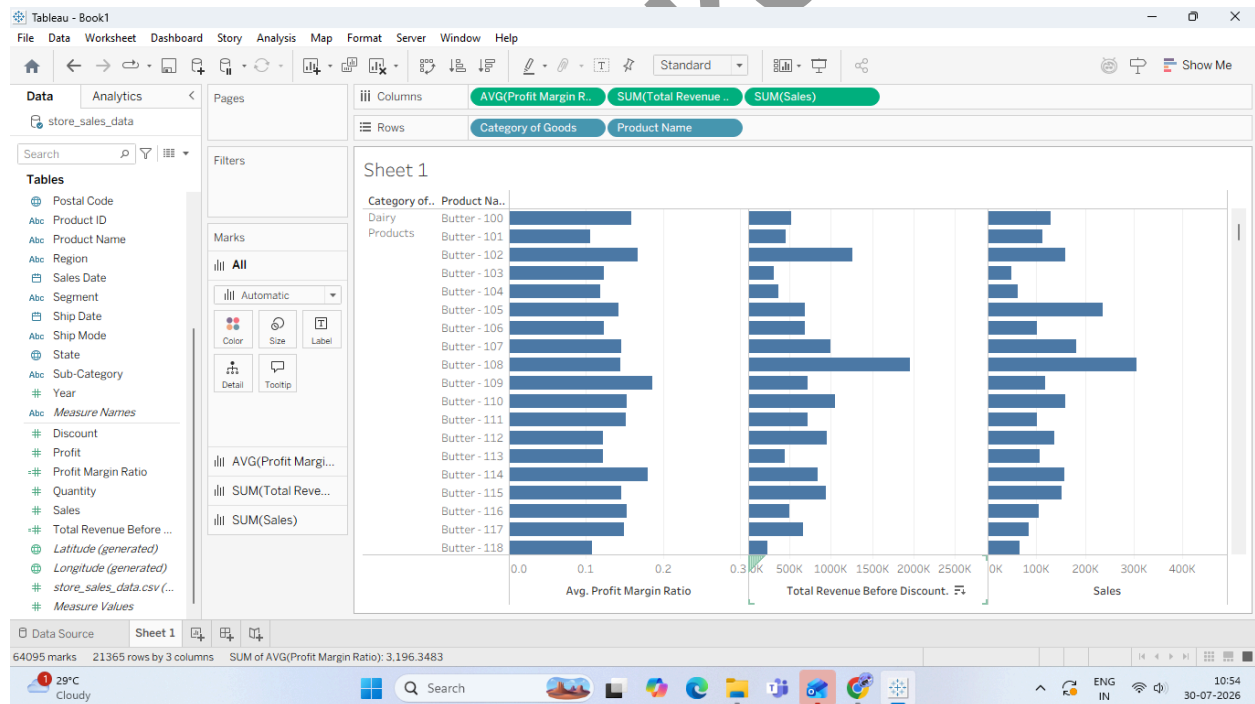
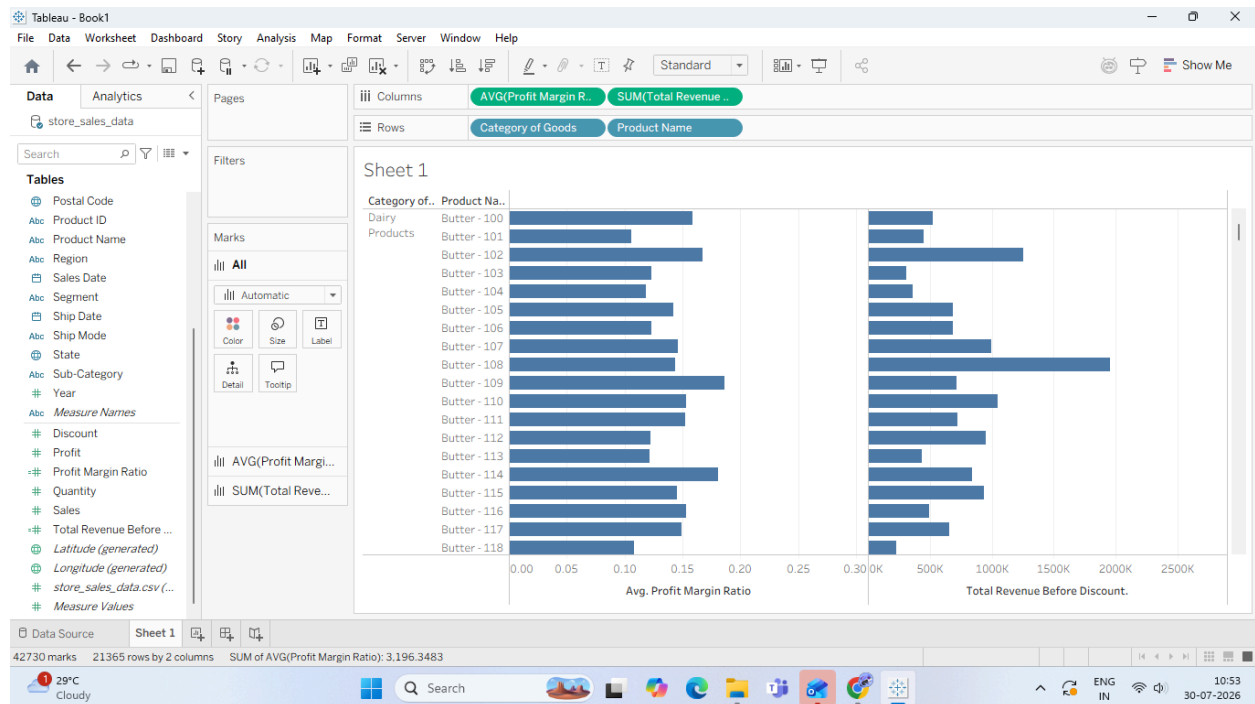
B: Performing Row-Level Calculations

1. Creating a Row-Level Calculated Field



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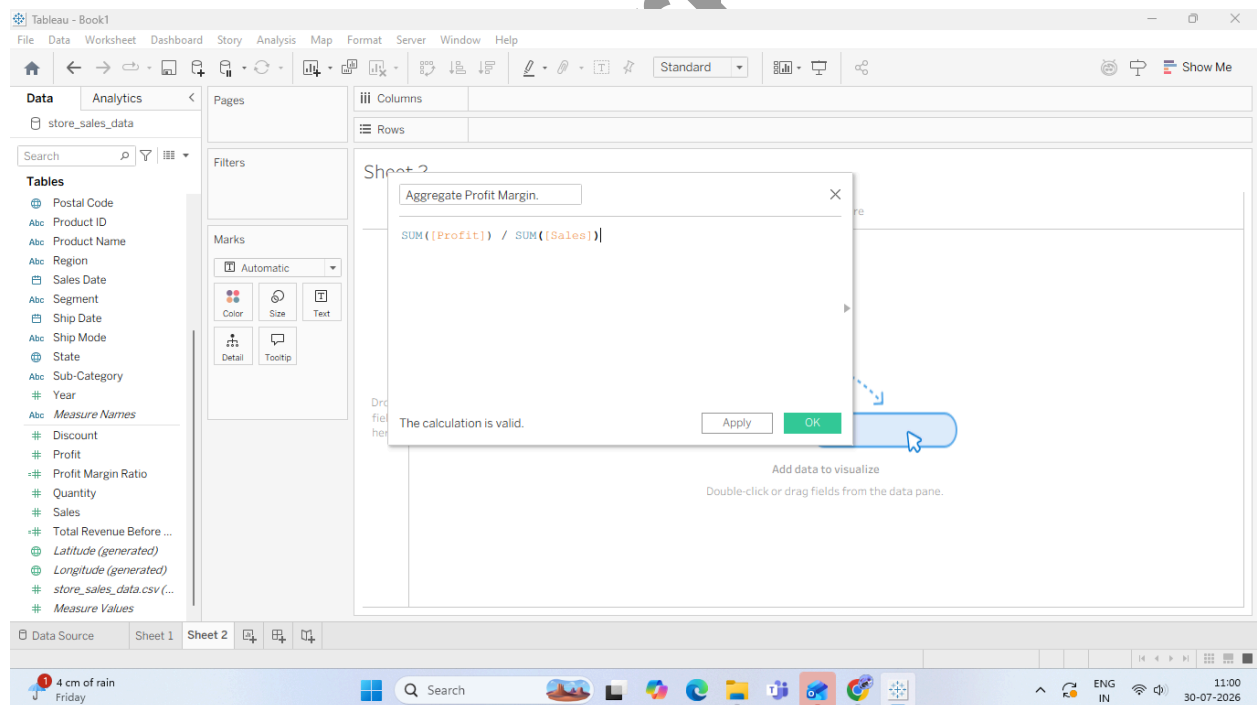
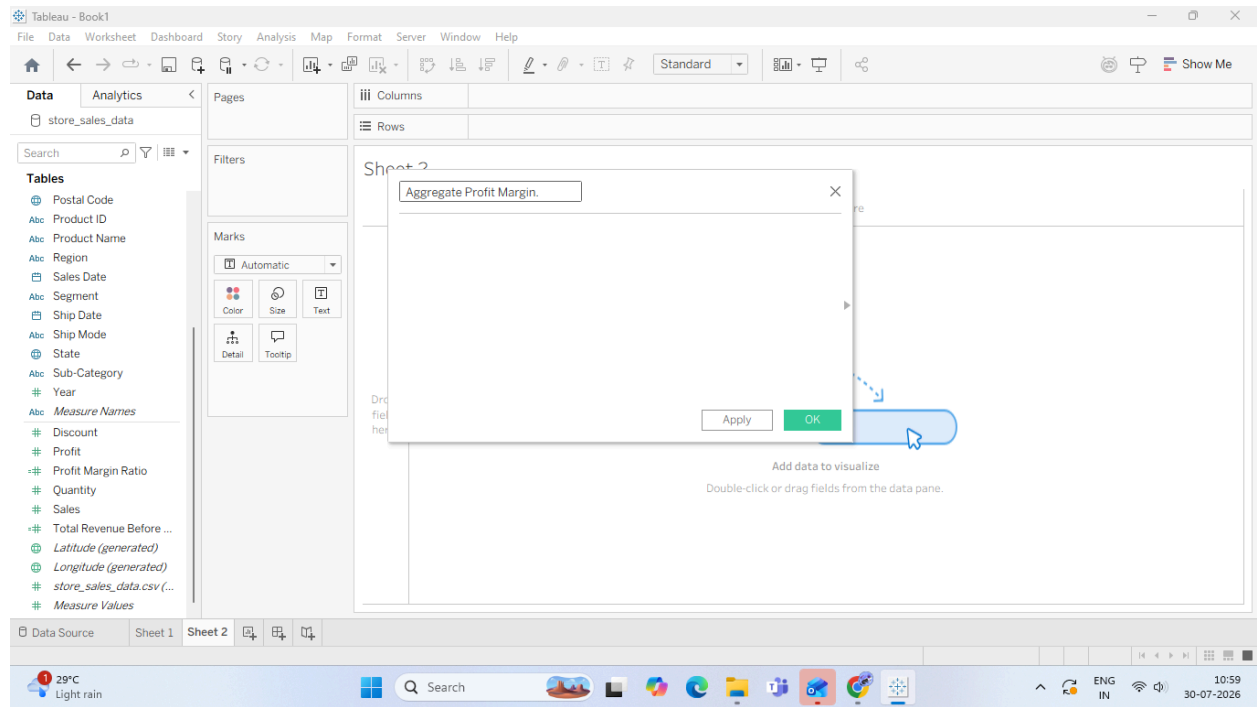
2. Putting the Field to Use



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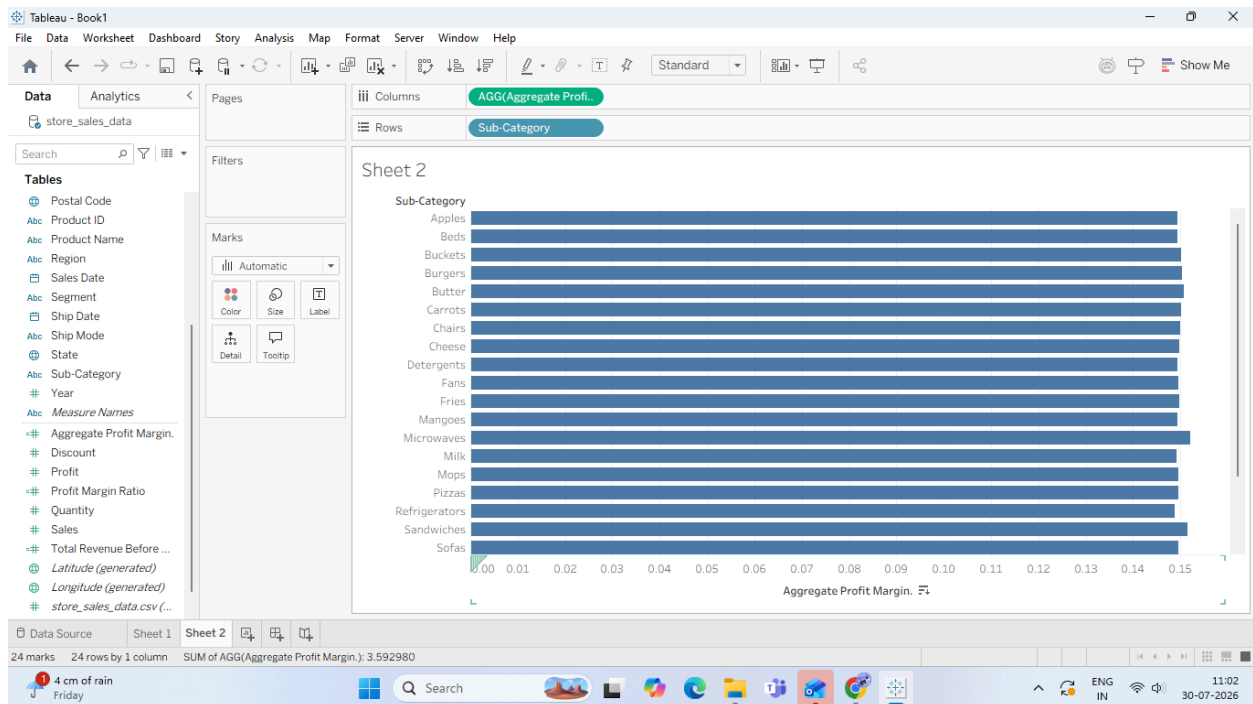
C: Performing Aggregate-Level Calculations

1. Creating an Aggregate Calculated Field



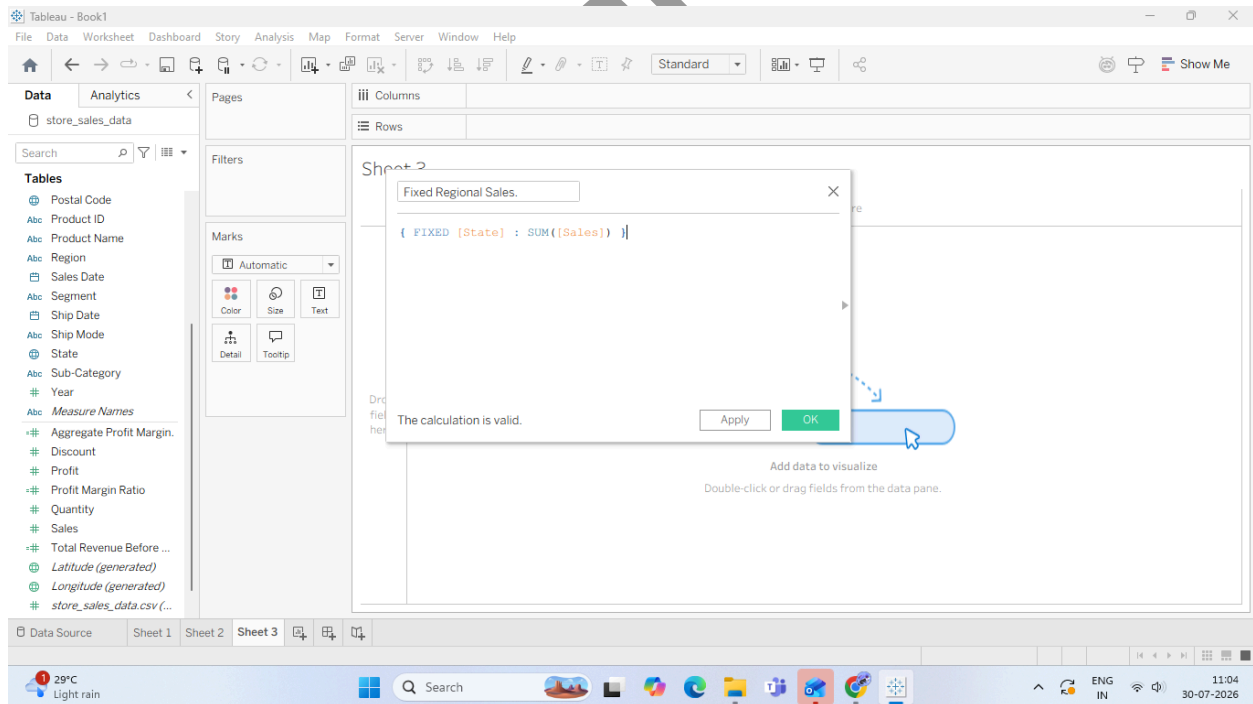
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2. Putting the Field to Use



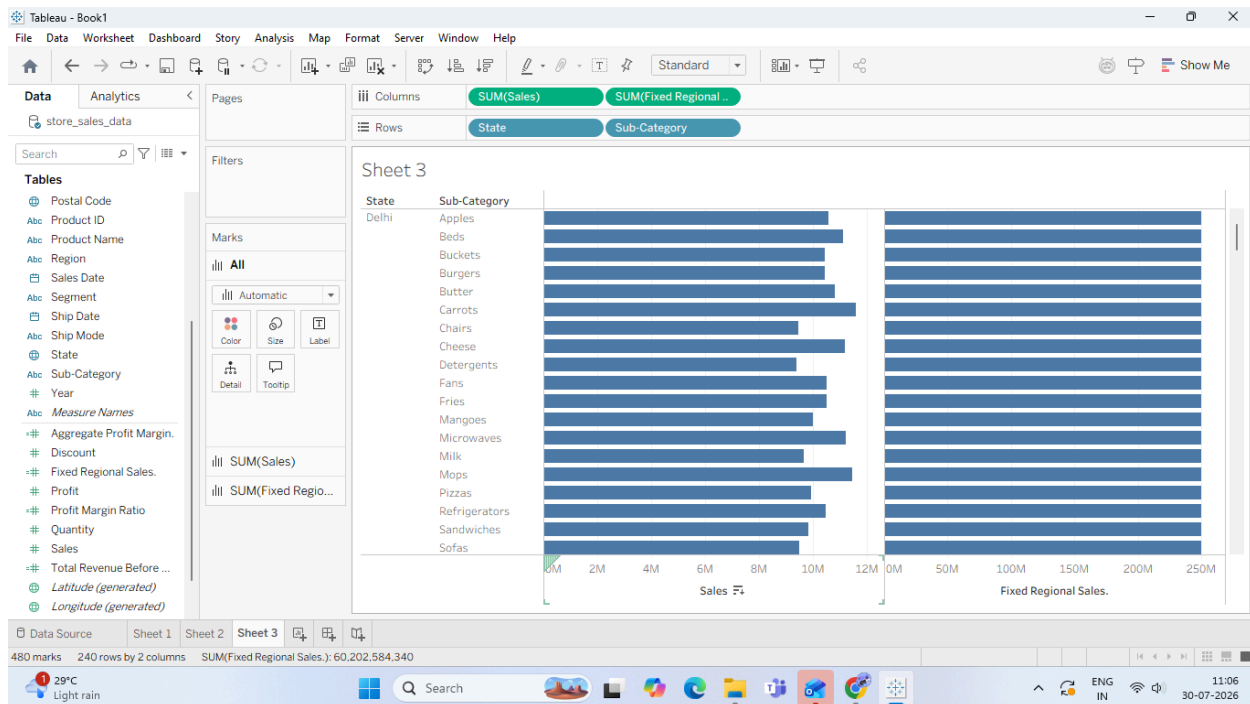
D: Implementing FIXED Level of Detail (LOD) Calculations

1. Creating a FIXED LOD Calculated Field



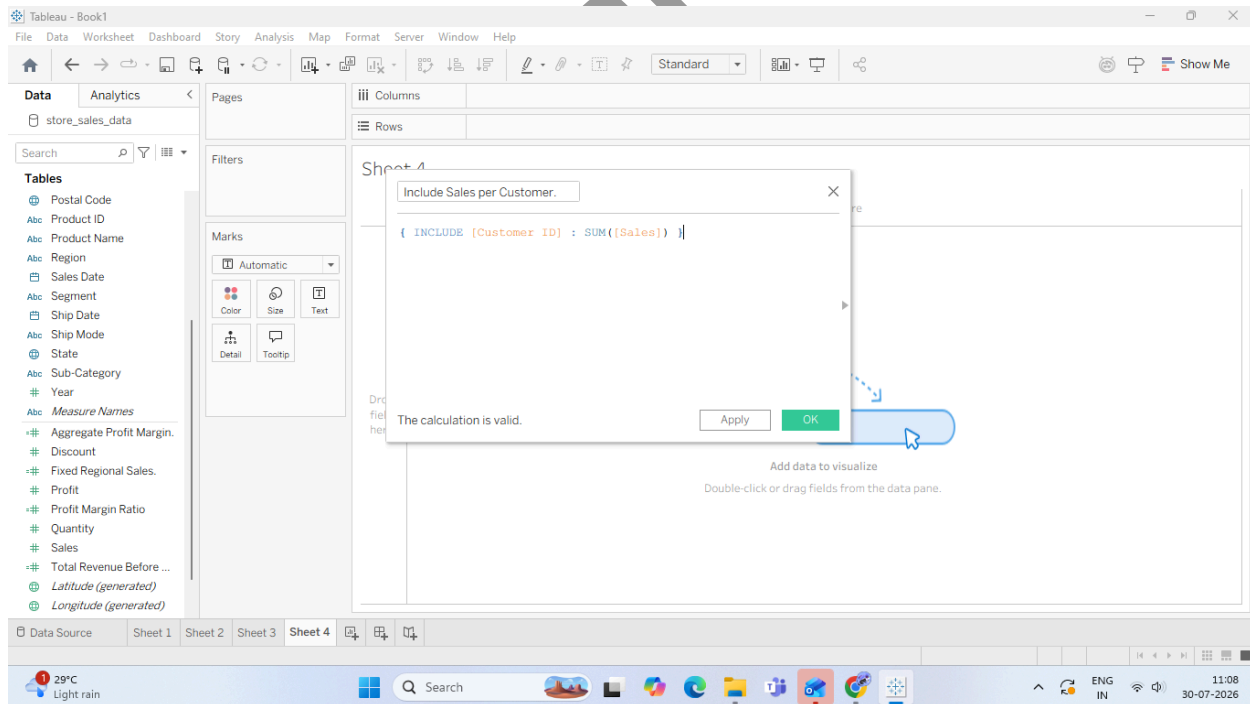
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2. Putting the Field to Use

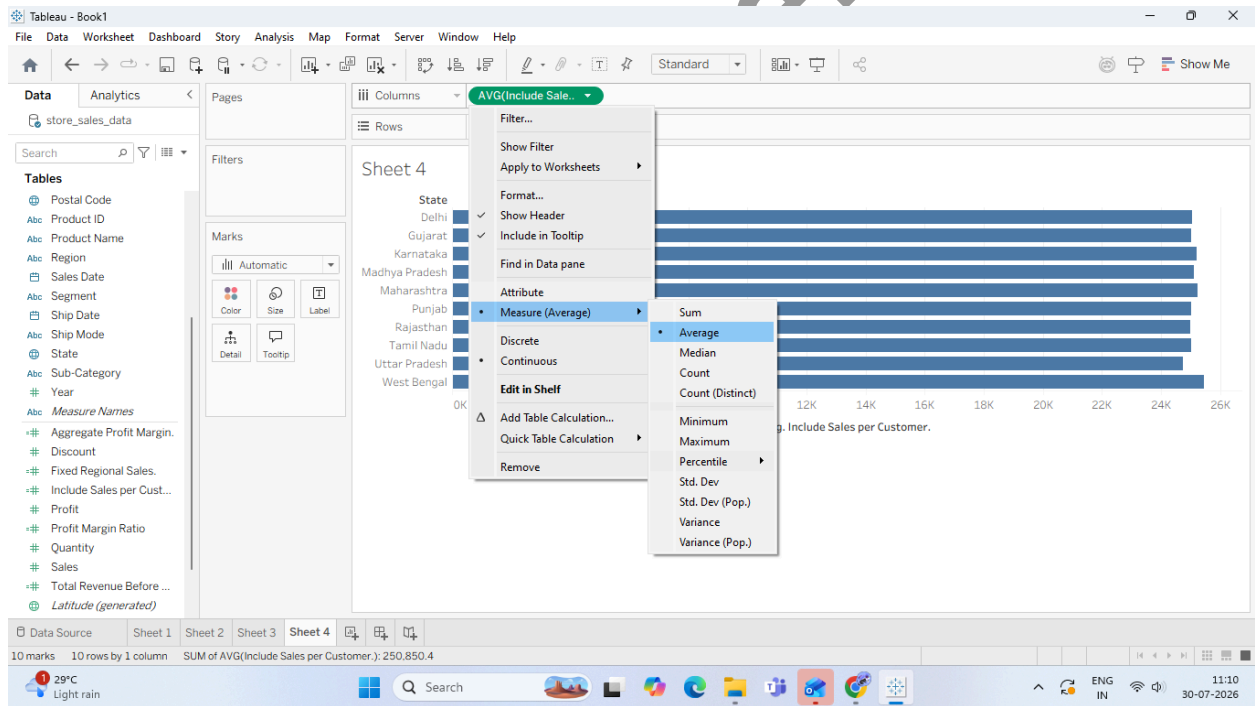
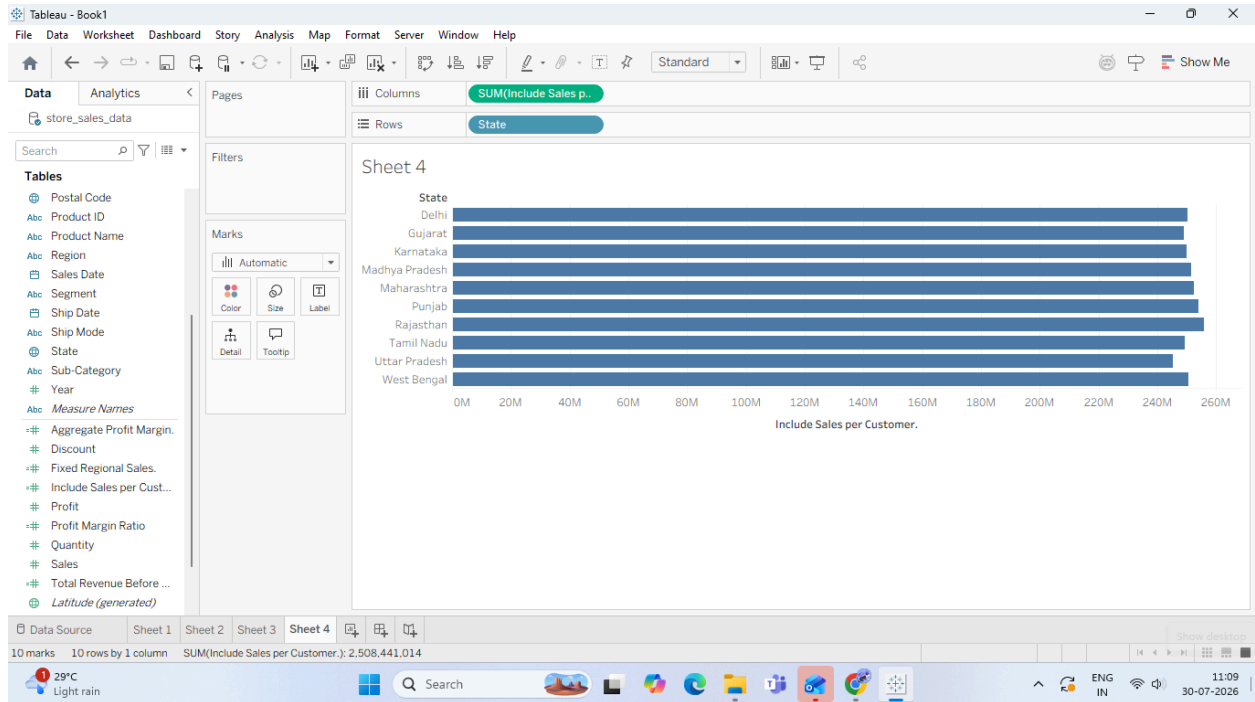


E: Using INCLUDE and EXCLUDE LOD Expressions

1. Creating an INCLUDE LOD Calculated Field



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3. Creating an EXCLUDE LOD Calculated Field

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Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Back Forward Refresh Save Print Export Copy Paste Undo Redo Standard Show Me

Data Analytics

store_sales_data

Search

Tables

- Category of Goods
- City Type
- Country
- Customer ID
- Customer Name
- Date of Birth
- Last Name
- Order Date
- Order ID
- Outlet Type
- Postal Code
- Product ID
- Product Name
- Region
- Sales Date
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Year
- Measure Names

Columns

Category of Goods Sub-Category

Rows

Sheet 5

Category of.. Sub-Category

Dairy	Butter	Abc
Products	Cheese	Abc
	Milk	Abc
	Yogurt	Abc
Electric	Fans	Abc
Appliances	Microwaves	Abc
	Refrigerators	Abc
	Washing Machines	Abc
Fast Food	Burgers	Abc
	Fries	Abc
	Pizzas	Abc
	Sandwiches	Abc
Furniture	Beds	Abc
	Chairs	Abc
	Sofas	Abc
	Tables	Abc
Household	Buckets	Abc
Items	Detergents	Abc
	Mops	Abc
	Utensils	Abc
Seasonal	Apples	Abc
Fruits &	Carrots	Abc

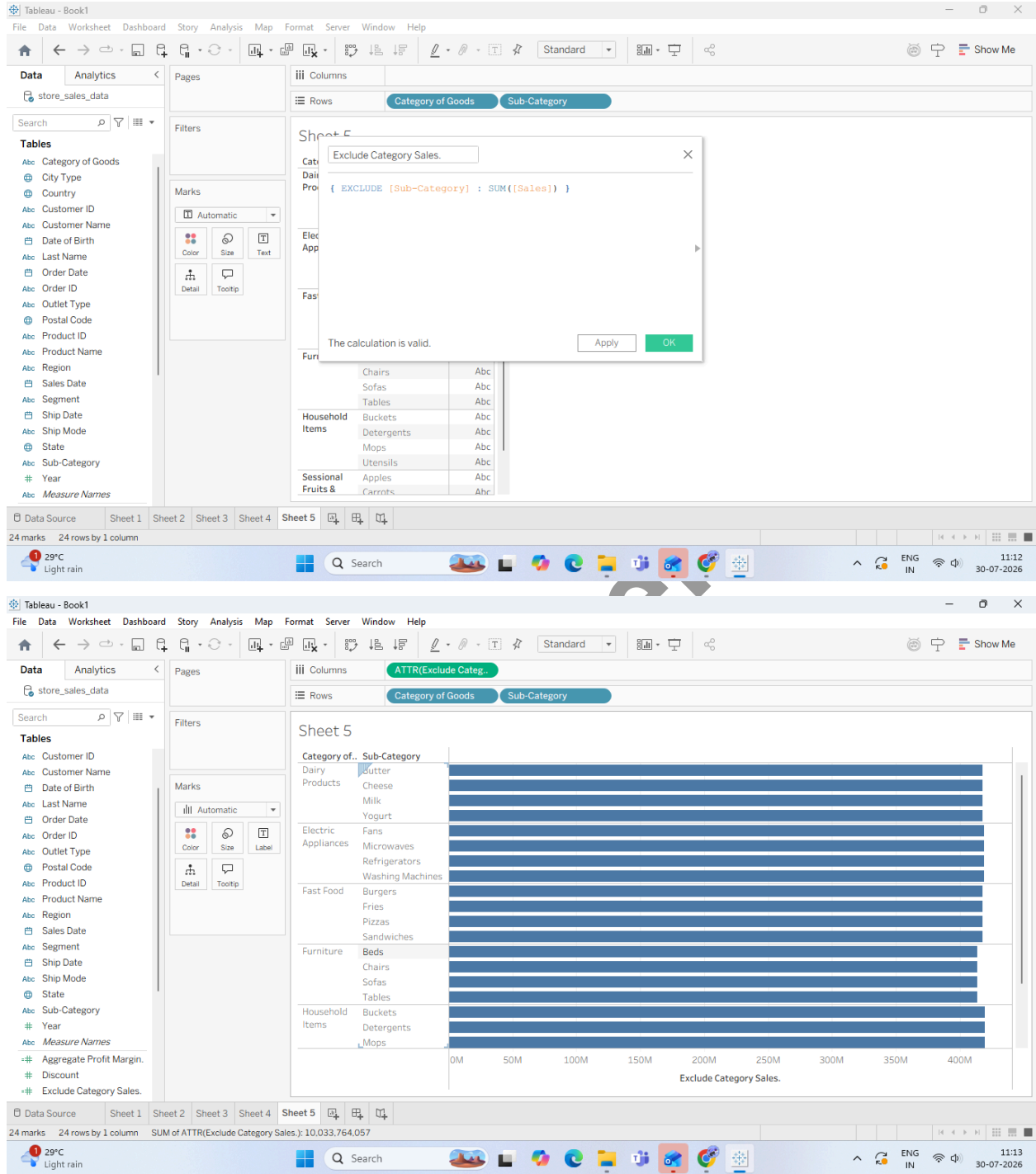
24 marks 24 rows by 1 column

29°C Light rain

Search

ENG IN 11:11 30-07-2026

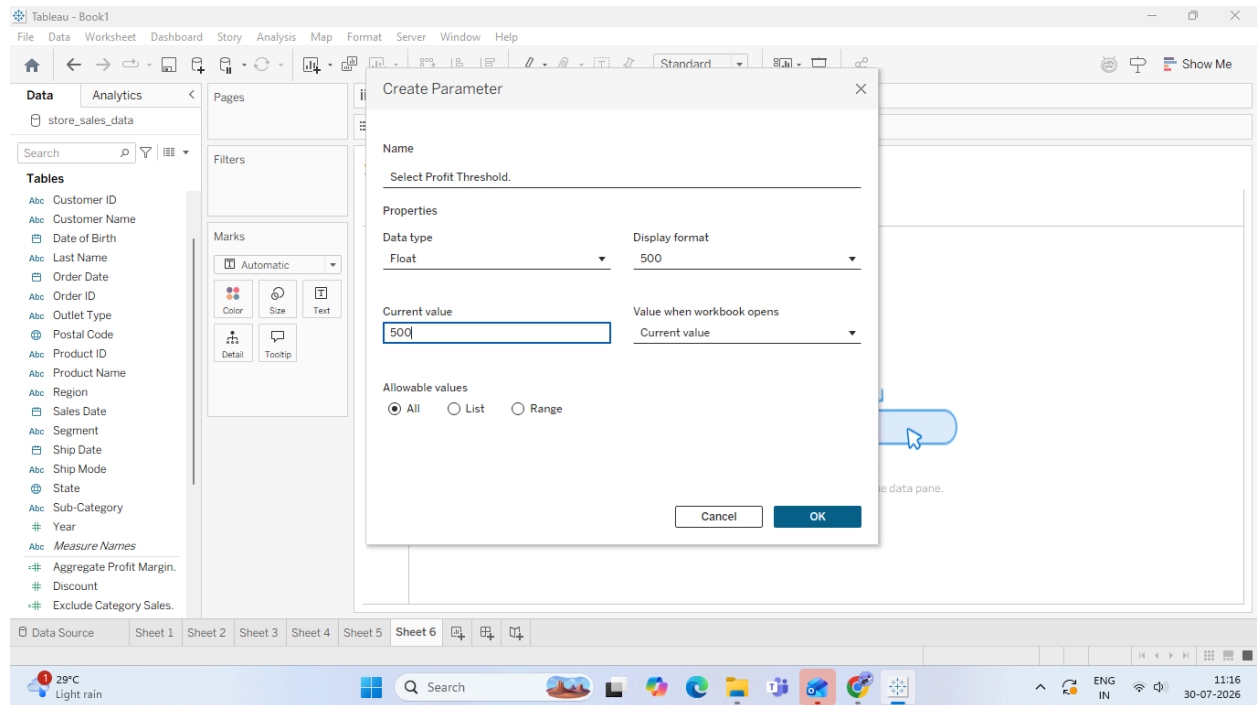
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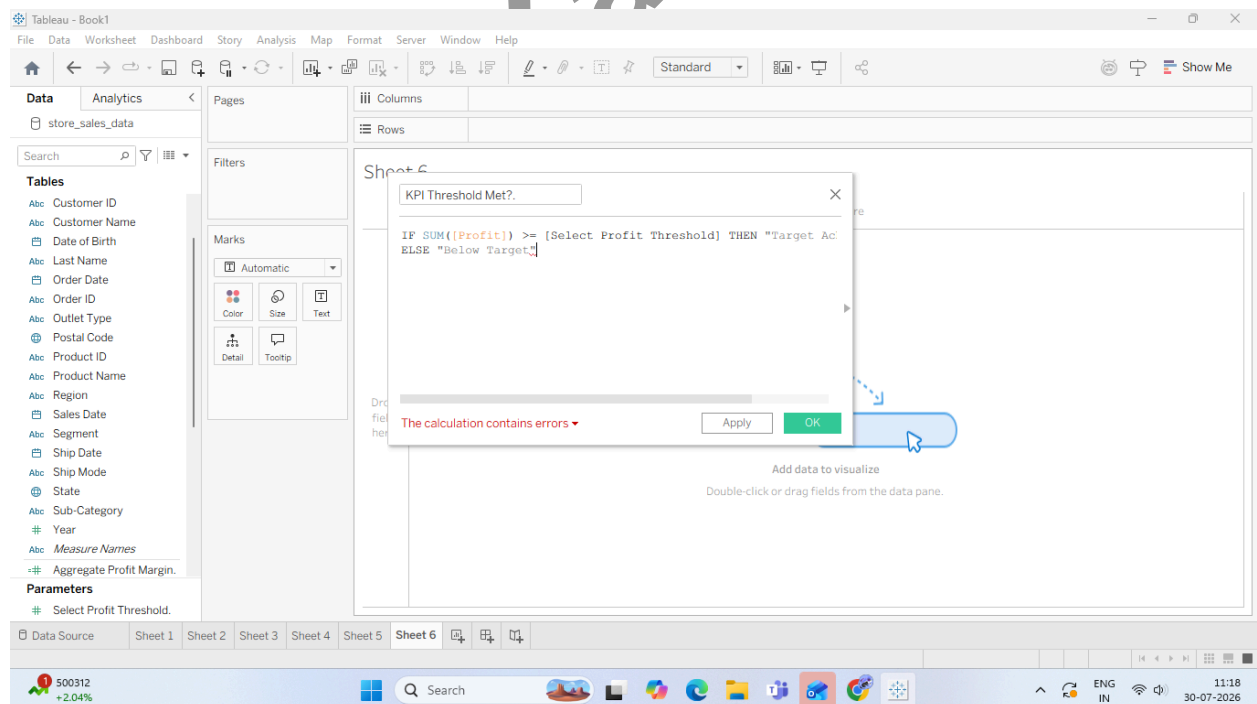
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F: Creating and Using Parameters for Dynamic Analysis

1. Creating a Parameter



2. Linking the Parameter to a Calculated Field



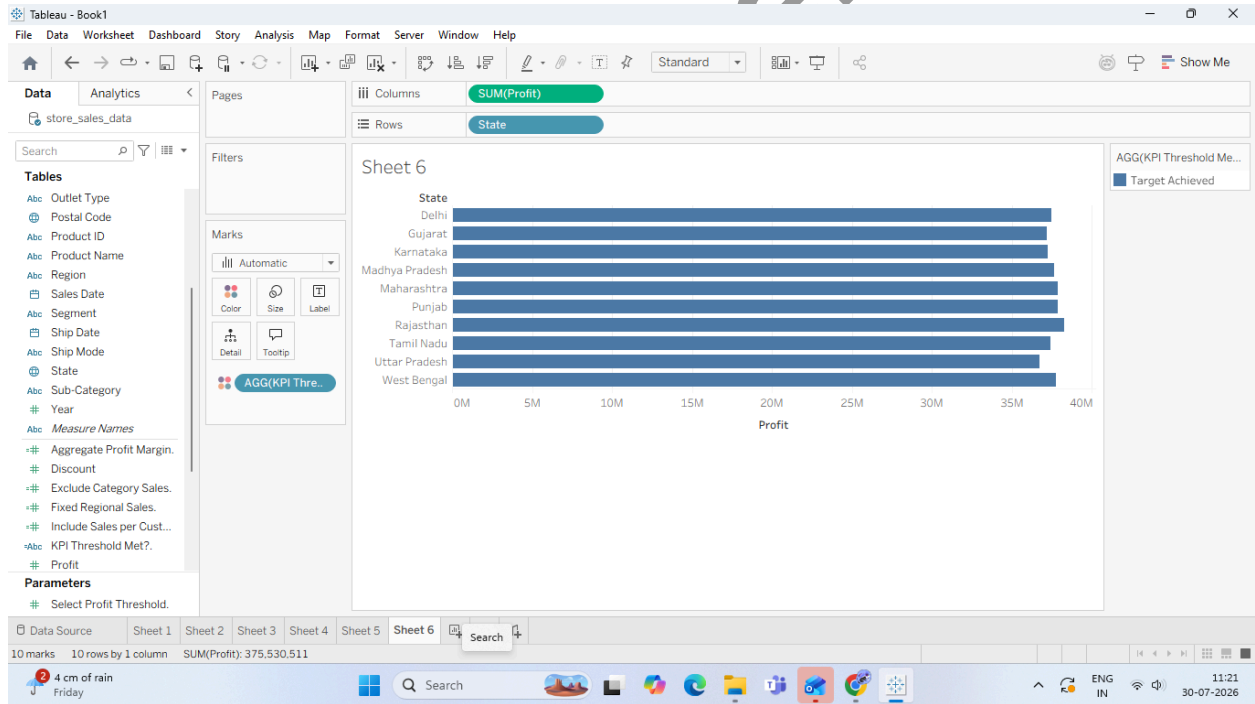
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The screenshot shows the Tableau Desktop interface. A dialog box titled 'KPI Threshold Met?' is open, displaying the following calculated field formula:

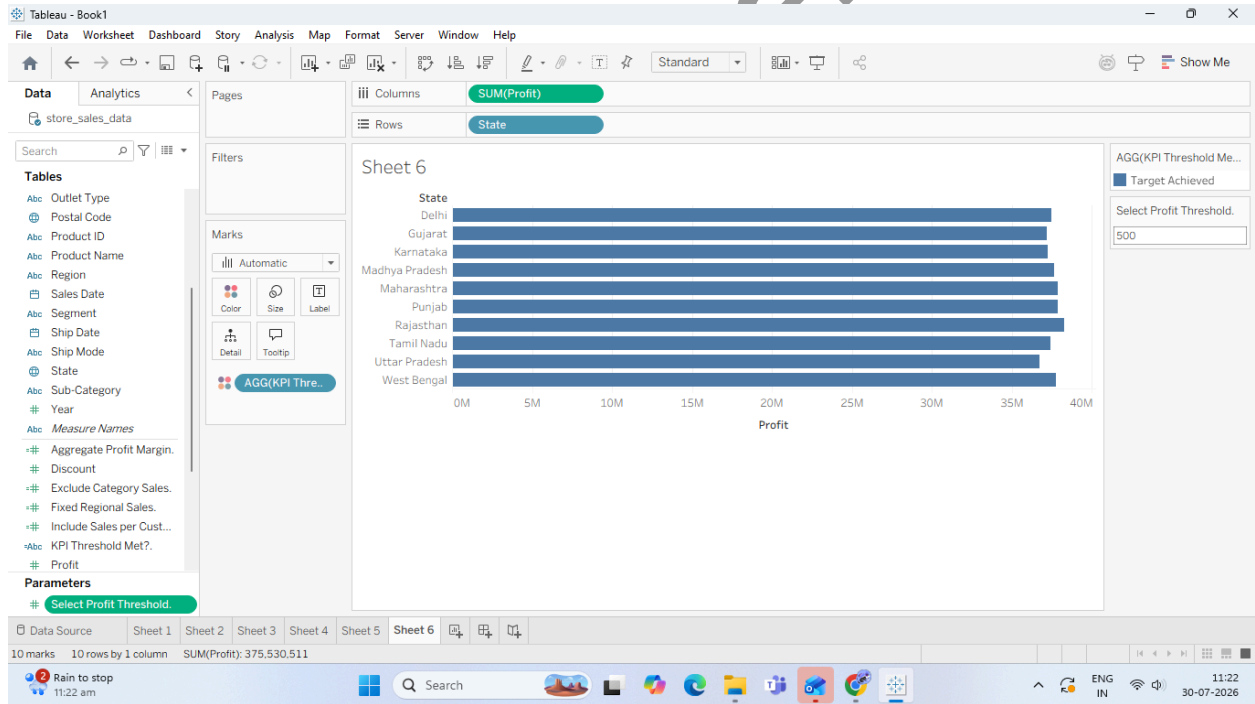
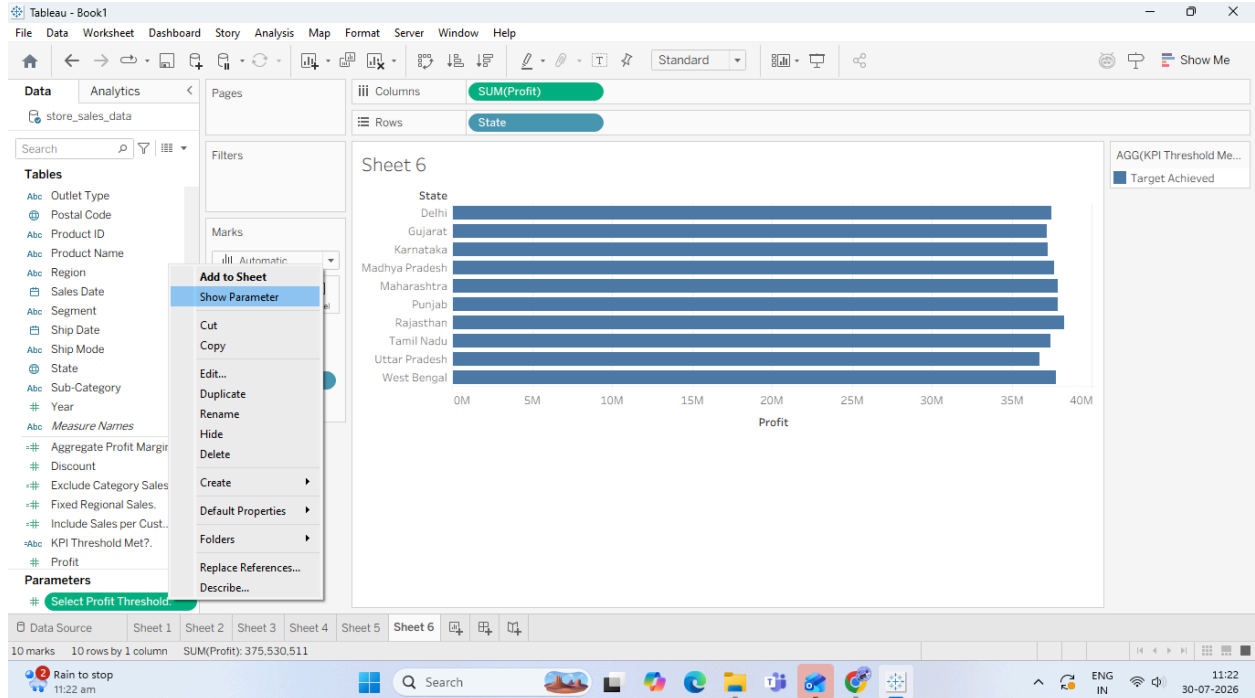
```
IF SUM([Profit]) >= [Select Profit Threshold.] THEN "Target Achieved"
ELSE "Below Target"
END
```

Below the formula, a message states 'The calculation is valid.' with 'Apply' and 'OK' buttons. A tooltip on the right side of the dialog box reads: 'Add data to visualize Double-click or drag fields from the data pane.'

The background interface shows the 'Data' pane with 'store_sales_data' as the data source. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. The 'Filters' shelf is empty. The 'Parameters' section shows 'Select Profit Threshold.' as a parameter.



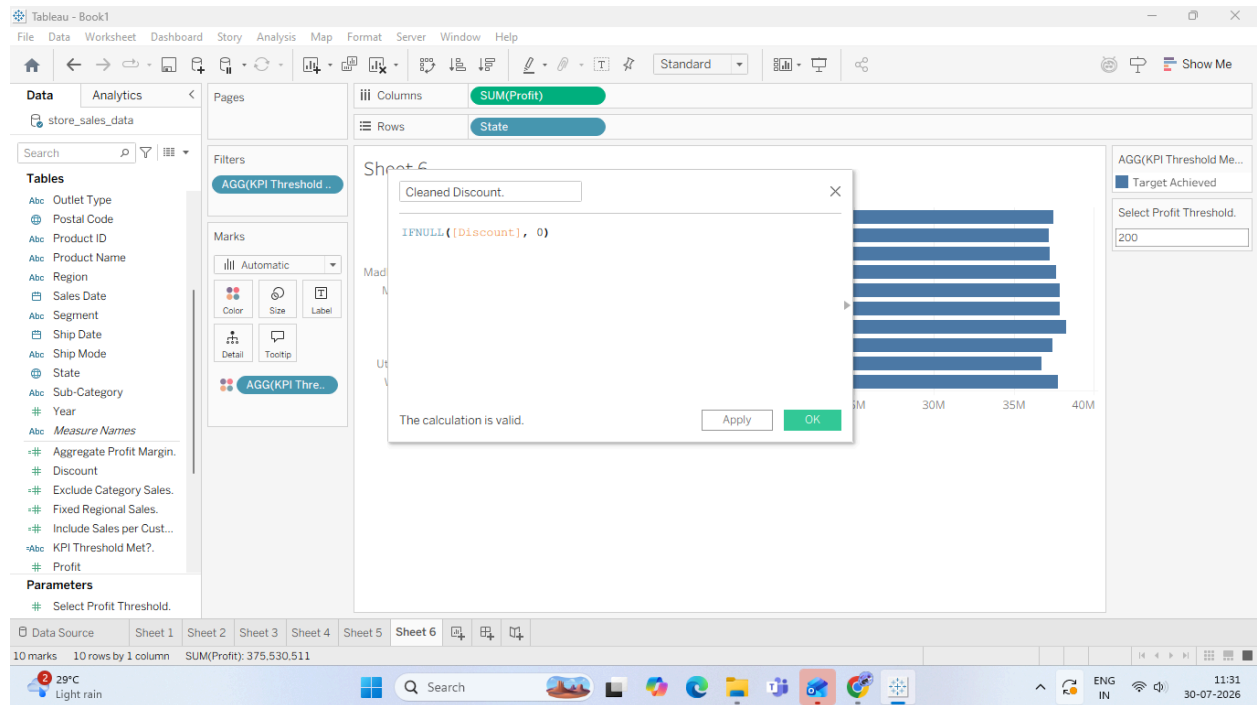
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G: Apply Calculations to Fix Data Issues and Enhance Insights

1. Handling Null or Missing Values with IFNULL



2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

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